

SEA4001W: Exercise 5 - Unconditional stability

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Exercise 5

$$C^{n+1} = \frac{C^n}{1 + K \Delta t} \rightarrow \text{euler backward scheme}$$

$$C^{n+1} = \underbrace{\frac{1}{1 + K \Delta t}}_{\text{amplification factor}} \times C^n$$

$$K > 0, \Delta t > 0$$

$$\therefore K \Delta t > 0$$

$$\therefore 1 + K \Delta t > 1 > 0$$

$$\therefore \frac{1}{1 + K \Delta t} > 0 \text{ always}$$

Since K & Δt are positive, the denominator $(1 + K \Delta t)$ is always greater than 1, making the amplification factor always positive, no matter how large Δt is.

$$1 + K \Delta t > 1 \longrightarrow \frac{1}{1 + K \Delta t} < 1$$

$$\therefore 0 < \frac{1}{1 + K \Delta t} < 1 \text{ for all } \Delta t > 0$$

for it to be stable, it must stay positive & not grow. Since the denominator is always greater than 1, the amplification factor is always a fraction between 0 & 1. Thus, each time step makes C smaller.

$$0 < \frac{1}{1 + K \Delta t} < 1 \text{ for all } K > 0, \Delta t > 0$$

$\therefore$ There is no restriction on Δt . The scheme is unconditionally stable. (the amplification factor is always between 0 & 1 regardless of the time step)